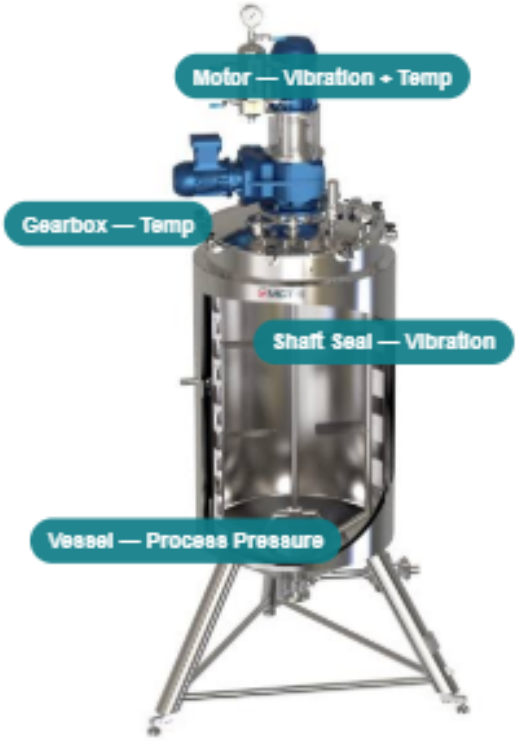
Preventive Maintenance of Agitator

Using IoT, Edge Gateway, PLC, OPC UA, Node-RED, MQTT and ThingsBoard Cloud



Document By : Vaishnavi Dadmal

Project Overview

Problem Statement

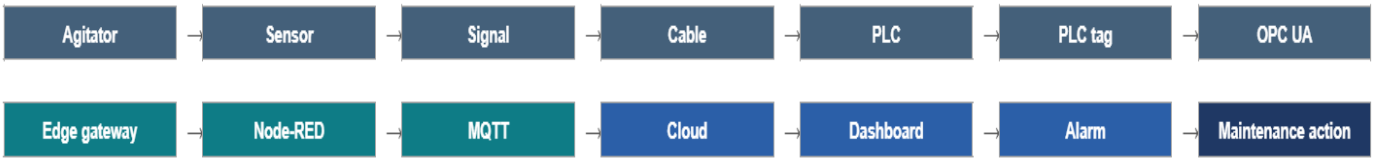
Agitator problems such as bearing wear, misalignment, imbalance, seal leakage or overload usually develop over time and change measurable quantities such as temperature, vibration and current. Without continuous measurement and history, these changes are noticed late, by operators or after a stoppage. There is no central, historical, remotely viewable record of agitator condition. Current maintenance practice at the plant: On site required (SCADA).

Need for Agitator Monitoring

- Unplanned downtime and batch impact when an agitator stops during processing.
- Secondary damage when a small fault is found late.
- Maintenance planning: knowing which machine needs attention first.
- Evidence: a stored trend and alarm history for root-cause review.

Project Objective

Continuously monitor important agitator parameters, identify abnormal conditions through threshold and rule-based logic, display the condition on a dashboard, generate alarms and support maintenance decisions.
Focus: preventive maintenance and condition-based maintenance.



Preventive Maintenance

Maintenance strategies differ in what triggers the work.

Strategy	Trigger	Example on an agitator	This project
Corrective	After failure	Replace seal after product leak	Not the goal: this is what the project tries to avoid
Preventive	Fixed time or running hours	Grease bearings every N hours; inspect coupling each shutdown	Supported: operating hours and inspection status shown on dashboard
Condition-based	Measured condition crosses a limit	Alarm when bearing temperature or vibration exceeds warning limit	Primary scope: live parameters, thresholds, NORMAL > WARNING > CRITICAL

Strategy	Trigger	Example on an agitator	This project
Predictive	Trend/model predicts remaining life	Rising 1x vibration trend predicts imbalance weeks ahead	Enabled by historical telemetry; trend rules are future scope, not claimed in prototype

IoT and IIoT concepts used in the project

Concept	What it is	In this project
IoT	Devices send measurements over a network to a remote system	Agitator values reach a cloud dashboard
IIoT	IoT for plant equipment: PLC data, industrial standards, reliability and security	Siemens PLC is the data source; control stays in the PLC
Sensor / transmitter	Converts a physical quantity to an electrical signal (4-20 mA, RTD, pulse, digital)	Temperature, vibration, current, pressure, speed
PLC + tag	Reads signals, scales to engineering units, stores in named tags	Source of all agitator data
OPC UA	Industrial data-access standard: server offers nodes, client reads or subscribes	PLC = server, Node-RED = client (prototype: Prosys)
Edge gateway	Plant-side computer that collects, validates, buffers and forwards data	Industrial gateway running Node-RED
Node-RED	Flow-based tool linking inputs, processing and outputs	OPC UA client, validation, mapping, MQTT OUT
MQTT	Light publish/subscribe messaging: broker, topic, payload, QoS, TLS	Publishes telemetry to ThingsBoard
Telemetry	Time-stamped measurement data sent remotely	JSON key/value, e.g. one value per parameter

Cloud computing concepts used in the project

Concept	What it is	In this project
Cloud computing	Storage and software on remote servers reached over the internet	Monitoring without a plant-side server
ThingsBoard Cloud	Hosted IoT platform	Receives MQTT telemetry

Device + access token	Digital twin of a thing; token is its secret identity	One device per agitator; token never shared
Time-series data	Values stored with timestamps	History and trend charts
Dashboard / widgets	Visual display of status, values, trends	Status, live values, trends, alarms
Alarm / rule engine	Rules raise Warning or Critical alarms with clear conditions	Vibration, temperature, pressure, speed alarms
Historical data	Stored telemetry queried over time	Trend review, base for later predictive work

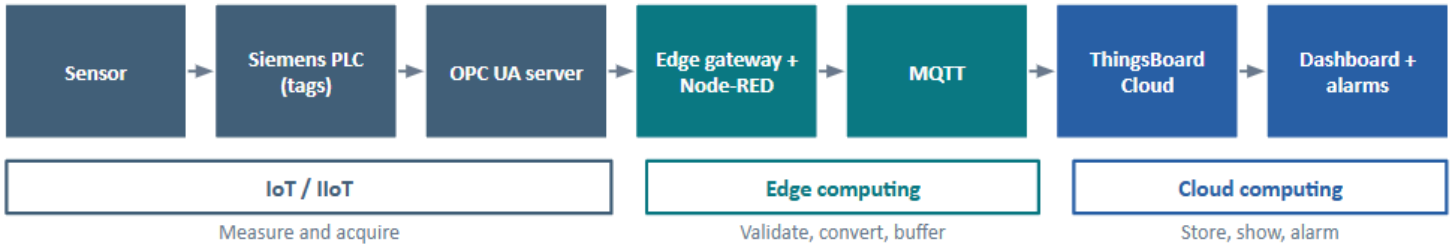
Advantages

- Remote access
- Central history
- Scalable
- Alarm notification

Limitations

- Needs internet
- Cybersecurity, privacy
- Latency, availability

How IoT and cloud fit together



STAYS LOCAL

- Control and interlocks
- Motor protection, E-stop
- Scaling and fault checks in PLC

GOES TO CLOUD (READ-ONLY)

- Telemetry, history, trends
- Alarms and notification
- Remote dashboard

Agitator Overview

Component	Function	Why it matters for monitoring
Motor	Provides torque	Current, temperature, power reveal load and electrical state
Gearbox	Reduces speed, increases torque	Oil/casing temperature, axial vibration
Coupling	Connects motor and gearbox/shaft	Misalignment shows as axial/radial vibration, coupling heat
Shaft	Transmits torque to impeller	Run-out and orbit indicate imbalance or bearing problems
Bearings	Support shaft	Temperature and vibration indicators
Mechanical seal / packed gland	Keeps product in the vessel	Barrier pressure/level, gland temperature, leakage
Barrier-fluid system	Pressurised fluid for seal (where applicable)	Pressure and level monitoring
Impeller	Moves product	Imbalance, erosion, build-up, blockage
Vessel	Holds product	Level and temperature influence cavitation and load
Support / bridge	Carries the drive	Rigidity affects vibration readings
VFD (if applicable)	Varies speed	Speed, fault feedback, VFD noise on cables, shaft voltage

Agitator Issues, Symptoms and Measurable Parameters

Issue	Possible cause	Observable symptom	Measurable parameter	Recommended sensor (source)	Signal
Bearing wear: dirt / contamination	Particle ingress, seal/breather defect	Noise, later heat and vibration rise	High-frequency vibration / acoustic emission	Vibration accelerometer + Acoustic Emission (AE)	Accel: 4-20 mA/comm.
Bearing wear: misassembly	Wrong fit, preload or installation	Heat soon after start, rough running	Bearing temperature, broadband vibration	Thermocouple + vibration sensor	RTD or 4-20 mA; vibration as above

Issue	Possible cause	Observable symptom	Measurable parameter	Recommended sensor (source)	Signal
Bearing wear: misalignment	Poor shaft/coupling alignment	Axial vibration, warm bearing	Axial 1x / 2x RPM amplitude	Triaxial accelerometer	4-20 mA / comm
Bearing wear: insufficient lubricant	Low grease/oil, wrong interval	Squeal, rising temperature	Crest factor, HF ultrasound, temperature	High-frequency ultrasound / crest factor + temperature	(crest factor computed from raw vibration; not available from a simple 4-20 mA transmitter)
Bearing wear: overloading	High torque, viscosity or process load	High current and bearing stress	Motor current, low-frequency envelope vibration, strain	Motor current (CT) + strain gauges	CT 4-20 mA; strain gauge
Bearing wear: corrosion	Moisture in oil/grease	Rust pits, noise bursts	Oil humidity, AE bursts	AE + oil humidity sensor	Dual Analog (2x 4–20 mA)
Bearing wear: improper journal finish	Surface finish out of tolerance	Shaft run-out, orbit deflection	Shaft displacement / orbit	Proximity probes (eddy current)	(probe + driver)
Bearing wear: VFD / electrical	Shaft voltage discharge through bearing	Fluting pattern, characteristic vibration	Shaft voltage, vibration frequency	Shaft voltage brush + vibration sensor	High-Frequency Voltage (AC/DC Peak-to-Peak) + Analog Dynamic Signal (IEPE / 4–20 mA)
Shaft misalignment	Angular or parallel offset, coupling wear	Axial shaking, thrust pounding, coupling heat	Axial/radial 1x-2x, phase across coupling, coupling temp, current, run-out	Triaxial accelerometer; two accelerometers across coupling; IR pyrometer; CT; proximity probes	4-20 mA/comm
Gearbox overheating	Lubricant breakdown, overload, misalignment	Hot casing, oil smell, noise	Oil temp, casing temp, motor amps, axial vibration	RTD PT100 oil probe; thermocouple + motor CT; triaxial accelerometer	RTD; 4-20 mA
Motor overload	Viscosity spike, process overload, supply issue	High current, hot motor, trips	Current, motor casing temp, VFD faults, voltage	CT; RTD/thermocouple on casing; VFD fault feedback; voltage monitoring + CT	4-20 mA; RTD; digital/comm

Issue	Possible cause	Observable symptom	Measurable parameter	Recommended sensor (source)	Signal
Cavitation	Low level, near-boiling batch	Gravel/marbles noise, erratic vibration	AE bursts, HF vibration, level, temperature	AE/ultrasonic sensor; accelerometer; level transmitter + temperature probe	4-20 mA; RTD
Blockage (impeller jam / build-up)	Solidified product, foreign object	Locked rotor, strong vibration, rubbing	Current, speed, 1x radial vibration, AE	VFD feedback / CT; radial accelerometer; accelerometer + AE	4-20 mA; comm
Impeller imbalance	Mass loss/gain, scaling, bent shaft	Radial vibration, wobble	1x radial amplitude, shaft orbit, current	Radial accelerometer; proximity probes; accelerometer + CT	4-20 mA
Impeller erosion	Cavitation or abrasive slurry	Reduced mixing efficiency, drift	AE, vibration and current trend over batches	AE; accelerometer + CT; radial accelerometer	4-20 mA
Seal leakage	Face separation, barrier loss, dry running	Drip/leak, squeal, gland heat	Shaft run-out, barrier level, barrier pressure, gland temp, AE	Proximity probes; level transmitter on barrier pot; pressure transmitter + flow switch; temp probe; AE/ultrasonic	4-20 mA; DI; RTD

Sensor Selection

#	Parameter	Agitator issue	Sensor	PLC input	Unit	Reason for selection
1	Bearing / gearbox oil / casing temperature	Bearing wear, gearbox overheating, misalignment (coupling), lubricant	RTD PT100 thermometer (source: TM111, 3-wire)	RTD module (Siemens RTD/AI, or AI	deg C	Source specifies RTD with matched-resistance extension cable; slow thermal trend suits RTD
2	Motor casing / winding temperature	Motor overload, thermal runaway	RTD PT100 on motor casing (source: RTD/thermocouple)	RTD / AI	deg C	Source lists RTD/thermocouple on motor casing
3	Vibration	Bearing wear, misalignment, imbalance, blockage, erosion, gearbox bearings	Vibration accelerometer; triaxial for alignment diagnosis (source)	AI (4-20 mA) or comm module	mm/s RMS or g	Most universal indicator for rotating faults. Only trend/amplitude, not diagnosis
4	Motor current	Overload, blockage, misalignment drag, bearing overloading	Current transducer (CT), 4-20 mA (source)	AI	A	Source lists CT for overload, locked rotor, binding
5	Speed (RPM)	Blockage (speed to zero), VFD/process check, 1x/2x frequency reference	Inductive proximity switch on shaft/coupling target (source: proximity switch, DI) or VFD speed feedback	DI with high-speed counter or AI/comm from VFD	rpm	Needed to relate vibration order peaks and detect stall
6	Pressure (barrier / seal pot, as justified)	Seal leakage	Pressure transmitter (source: PMP50, 4-20 mA / HART)	AI	bar	Source rule: barrier pressure should be 1-2 bar above vessel pressure
7	Motor power and energy	Overload trend, energy dashboard, process load	Multifunction energy meter or VFD drive data	Comm via PLC network, or AI/pulse	kW, kWh	Dashboard requires motor power and energy; source CT alone gives current only

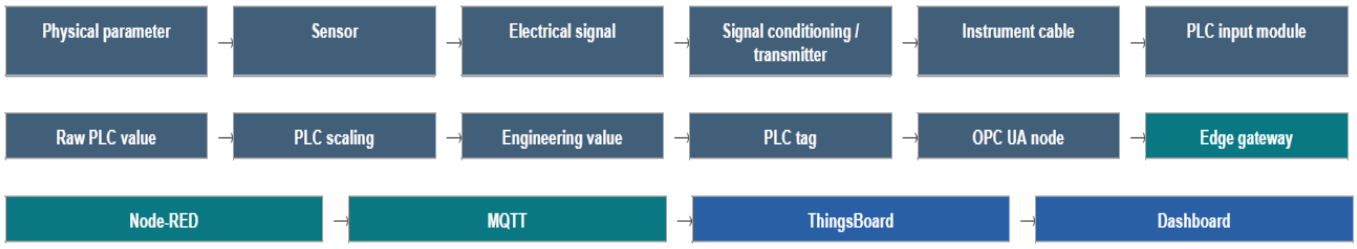
#	Parameter	Agitator issue	Sensor	PLC input	Unit	Reason for selection
8	Barrier pot level	Seal leakage (barrier fluid loss)	Level transmitter on barrier pot (source: guided radar FMP51 / ultrasonic / level transmitter 4-20 mA)	AI	% or mm	Source lists level transmitter for barrier fluid loss
9	Shaft run-out / orbit	Imbalance, seal leakage, journal finish	Non-contact eddy-current proximity probes (source)	AI or monitor via comm	um	Source lists for orbit/run-out
10	Acoustic emission / ultrasonic	Contamination, cavitation, corrosion, dry running, erosion	AE / ultrasonic sensor (source)	Analog	Db	Source gives early detection before heat or vibration
11	Run status / fault	Equipment status, repeated trips	Contactor/VFD auxiliary contacts, VFD fault feedback	DI	bool	Source: agitator is 1 DI + 1 DO; VFD fault feedback listed under motor overload

Cable Selection

Instrument	Signal	Cable type	Cores	Conductor size (sq.mm)	Shield	Approx. distance
Temperature transmitter (RTD/TC, head-mounted)	4-20 mA	Shielded twisted pair (instrument cable)	2	0.75-1.5	Individual pair or overall shield	Up to 300 m
RTD thermometer (TM111, direct to PLC)	RTD 3-wire	RTD extension cable (matched resistance)	3	0.5-1.0	Yes	Up to 100 m; longer needs transmitter at head (source)
Vibration sensor (accelerometer)	Analog 4-20 mA / comm	Shielded coaxial or twisted pair	2	0.2-0.5	Yes (shielded)	Up to 100 m
Current transducer (CT)	4-20 mA	Shielded twisted pair (instrument cable)	2	0.75-1.5	Yes	Up to 300 m

Instrument	Signal	Cable type	Cores	Conductor size (sq.mm)	Shield	Approx. distance
Pressure transmitter (PMP50)	4-20 mA / HART	Shielded twisted pair (instrument cable)	2	0.75-1.5	Yes	Up to 300 m
Level transmitter (FMP51 / FMR62B or equivalent)	4-20 mA / HART / comm	Shielded twisted pair (instrument cable)	2	0.75-1.5	Yes	Up to 300 m
Proximity switch (inductive)	Digital	Multi-core control cable (unshielded, source)	3	0.75-1.5	No (Unshielded Control Cable)	Up to 100 m
Encoder (incremental), if used for rpm	Digital pulse	Shielded twisted pair (pulse cable)	4-6	0.5-0.75	Yes	Up to 50 m
Energy meter / digital meter	Comm	Shielded twisted pair or Cat5e/Cat6	2-4	0.2-0.75	Yes	Up to 100 m
Agitator motor (DOL via contactor)	Digital control; power	Multi-core control cable + armoured power cable	Control 4; power 4	2.5-4 (source range)	Control unshielded (source)	Power up to 100 m; control up to 100 m
VFD-controlled motor, if applicable	Analog + digital	VFD-rated shielded symmetrical cable	4	2.5-6 (source range)	Yes	Up to 50 m unshielded / 150 m+ shielded (source)
PLC <-> Edge Gateway (Ethernet/PROFINET)	Ethernet	Shielded Cat5e/Cat6 (source) or industrial PROFINET Ethernet cable	4-8 core	0.2-0.5	Yes	Up to 100 m copper (Ethernet limit); fibre beyond (source: fibre pair for km range)

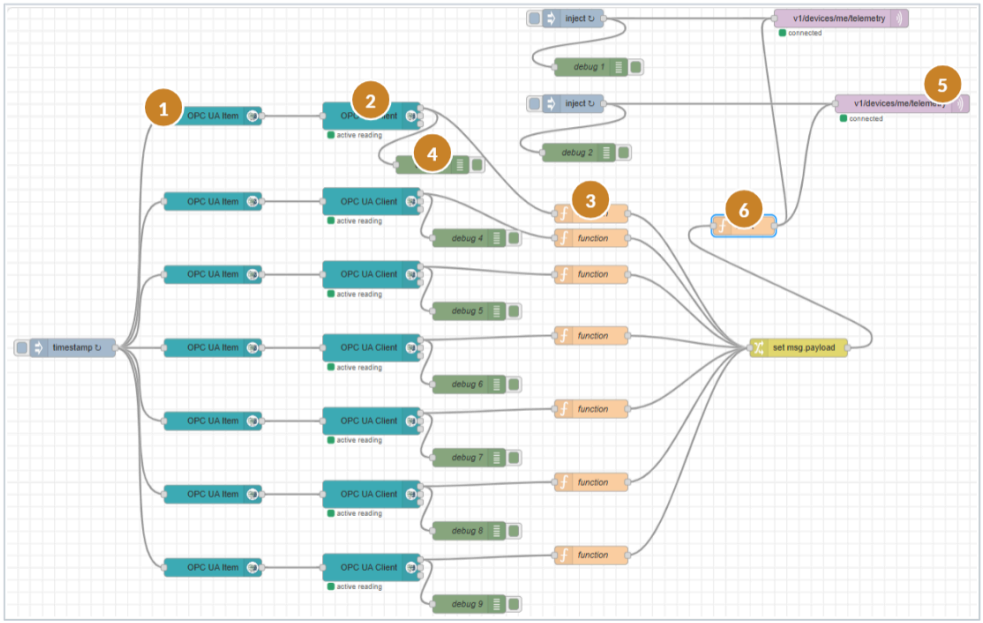
Data Acquisition Chain



Prototype

Prosys OPC UA Simulation Server → Node-RED OPC UA Client → Data processing → MQTT OUT → ThingsBoard Cloud → Dashboard / Alerts

Actual prototype: Node-RED flow



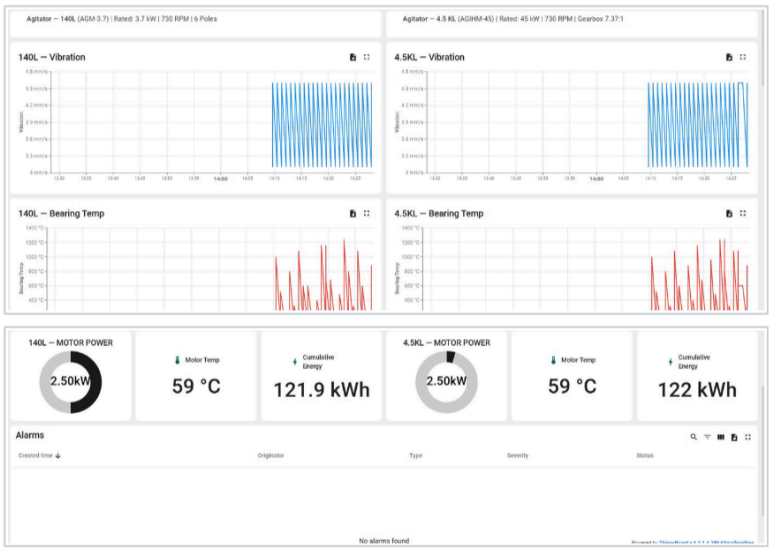
PROTOTYPE: SIMULATED DATA

1. **OPC UA Item**
   a simulated variable selected by NodeId
2. **OPC UA Client**
   reads the Prosys server ("active reading")
3. **Function nodes**
   per-signal data processing
4. **Debug**
   live message inspection
5. **MQTT output**
   v1/devices/me/telemetry, connected
6. **Alert function**
   alert processing before publishing

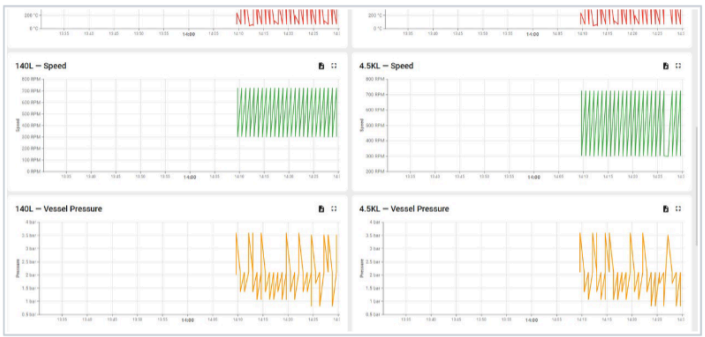
7 simulated OPC UA data paths

Actual prototype: ThingsBoard dashboard

Agitator condition monitoring: Agitator 140L and Agitator 4.5KL



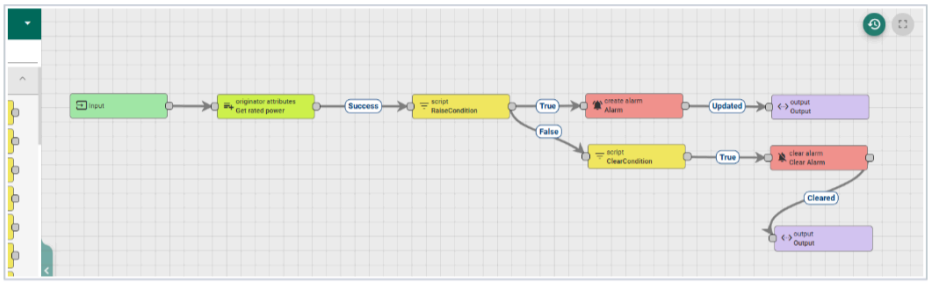
PROTOTYPE: SIMULATED DATA



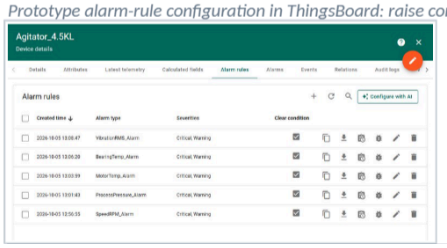
- Telemetry reaches ThingsBoard and is trended per agitator
- Tiles for motor power, motor temperature, energy
- Alarm table (empty at capture)

Actual prototype: ThingsBoard alarm rules

PROTOTYPE CONFIGURATION



Prototype alarm-rule configuration in ThingsBoard: raise condition, create alarm, clear condition, clear alarm.



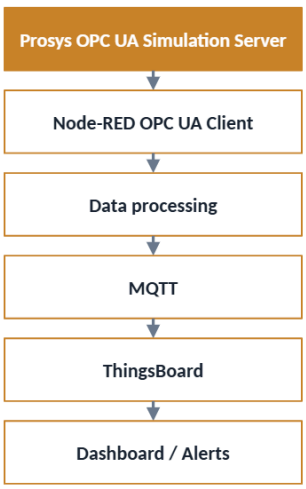
- VibrationRMS_Alarm
- BearingTemp_Alarm
- MotorTemp_Alarm
- ProcessPressure_Alarm
- SpeedRPM_Alarm
- Severities: Warning, Critical

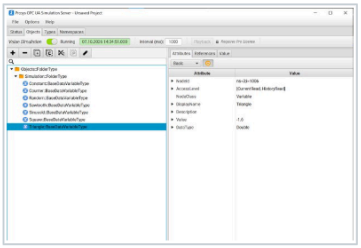


- Delay: condition must persist before raising
- Hysteresis: clears at a different level
- Clear condition: shown in rule chain
- Delay and hysteresis values: Information Required from Site/Company

Prototype vs actual industrial implementation

CURRENT PROTOTYPE (SIMULATED)



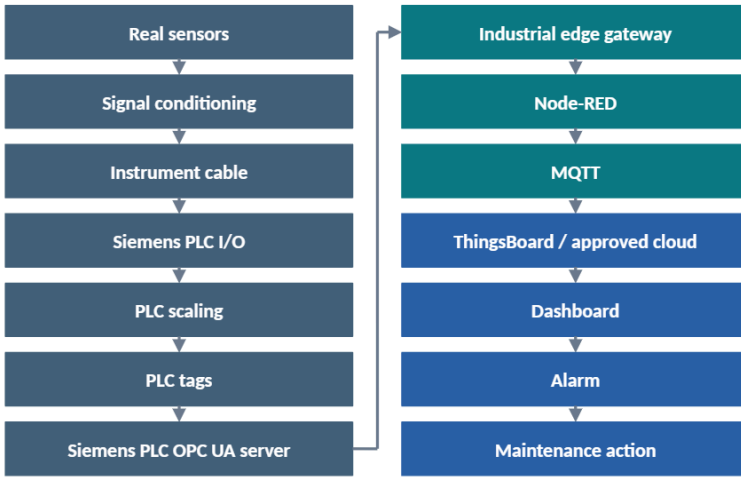


Prosys OPC UA Simulation Server: simulated OPC UA data source used for prototype validation.

7 simulated readings

Prosys is a simulator, NOT the Siemens PLC

ACTUAL INDUSTRIAL IMPLEMENTATION (FUTURE)



What the Prototype Proves

- OPC UA data acquisition by a client from an OPC UA server.
- Node-RED processing of the values.
- MQTT communication to ThingsBoard using a device access token.
- Cloud telemetry reception and storage.
- Dashboard visualization of status, trends and tiles.

- Alarm logic: ThingsBoard alarm rules with Warning and Critical severities and a clear condition.

Node-RED



Node-RED is the edge data-processing layer. It reads OPC UA values, validates and formats them and publishes telemetry. It does not replace the PLC and does not control the machine.

Item	Configuration	Prototype	Plant
Endpoint	OPC UA server URL	Prosys endpoint	PLC endpoint
Security	Policy and mode	None acceptable	Sign and Encrypt, trusted certificates
Authentication	Anonymous/user/certificate	Anonymous	Read-only user
Items	Node IDs to subscribe	Simulation nodes (browse to confirm)	Real PLC Node IDs
Subscription	Sampling and publishing intervals	e.g. 1 s (Example)	Per signal
Connection status	Status output / catch node	Show in debug	Raise comm-failure telemetry
Error handling	Catch node, retry, last-good-value	Basic	Buffer and retry

MQTT

MQTT is a small, efficient publish/subscribe protocol suited to devices on limited or unreliable links. The publisher (Node-RED) sends a payload to a topic on a broker (ThingsBoard); subscribers (ThingsBoard device) receive it.

MQTT term	Meaning	In this project
Broker	Server that receives and distributes messages	ThingsBoard MQTT endpoint (server URL)
Publisher / subscriber	Sends / receives messages by topic	Node-RED publishes; ThingsBoard device ingests
Topic	Hierarchical message address	v1/devices/me/telemetry (ThingsBoard device API)

MQTT term	Meaning	In this project
Payload	Message content	JSON key/value, optional timestamp
QoS	0 at most once, 1 at least once, 2 exactly once	QoS 1 is a sensible default; verify support
Keepalive	Heartbeat interval to detect dead connections	e.g. 60 s (EXAMPLE)
Retained message	Broker keeps last message for new subscribers	Not normally needed for telemetry
Authentication	Credentials at connect	ThingsBoard: access token as MQTT username, empty password
TLS	Encryption of the connection	Use TLS port in plant; port and certificate

ThingsBoard / Cloud Dashboard

Concept	Meaning	Use here
Cloud computing	On-demand compute/storage over the internet	Remote monitoring without plant-side servers
ThingsBoard Cloud	Hosted IoT platform	Device ingestion, dashboards, alarms
Device	Digital representation of a connected thing	One device per agitator (AGT_140L, AGT_4K5L, EXAMPLE names)
Device profile	Shared settings: transport, alarm rules	Alarm rules defined once for agitator devices
Access token	Secret credential for a device	Used as MQTT username. Never publish or screenshot it
Telemetry	Time-stamped measurements	All dynamic values (vibration, temperature, etc.)
Attributes	Relatively static metadata	Model, motor kW, location, maintenance status
Time-series data	Values stored with timestamps	Trend widgets, history
Widgets / Dashboard	Visual components arranged on a page	Section 14
Alarm	Stateful event with severity, ack and clear	Section 15
Rule Engine	Event-processing pipeline	Notifications, routing
Historical data	Stored telemetry queried over time	Trend analysis, maintenance records

Plant Implementation

The prototype is converted to a plant system by replacing the simulator with real sensors and the Siemens PLC. Each stage below is explained in practical commissioning terms.



Layer	Hardware	Software / protocol	Input > Output	Responsibility	Failure mode	Cybersecurity
1. Agitator	Motor, gearbox, bearings, seal, impeller	n/a	Process load > physical effects (heat, vibration, current)	Source of all symptoms	Mechanical faults per Section 6	Physical access control
2. Sensors / transmitters	PT100, accelerometer, CT, pressure transmitter, proximity (Section 7)	Analog / RTD / pulse	Physical quantity > electrical signal	Measure	Drift, damage, wrong mounting, loss of power	Tamper-evident enclosures
3. Cable and JB	Instrument cable, junction boxes, glands	n/a	Signal > signal	Carry signal with low noise	Open/short, noise pickup, water ingress	n/a
4. PLC I/O	Siemens AI / RTD / DI / HSC modules	PLC program	Signal > raw count > scaled value > tag	Scale, filter, range-check, fault-flag	Wire break, scaling error, module fault	Program protection, access levels
5. Industrial network	Switch, shielded Ethernet cable	Ethernet / PROFINET (as used by plant)	Frames	Transport	Link loss, loop, congestion	Segmentation, VLAN
6. PLC OPC UA server	PLC CPU (**SVR** model/FW/ licence)	OPC UA	Tags > nodes	Expose selected tags read-only	Server disabled, cert untrusted, node changed	Certificates, user auth, signed+encrypted policy

Layer	Hardware	Software / protocol	Input > Output	Responsibility	Failure mode	Cybersecurity
7. Edge gateway	Industrial PC/gateway	OS, Node-RED, OPC UA client	Node values > validated values	Collect, validate, buffer, publish	Reboot, flow stopped, disk full	Hardening, firewall, patching, least privilege
8. MQTT	Network to internet	MQTT over TLS	JSON telemetry > broker	Reliable lightweight transport	Broker unreachable, wrong token	TLS, token secrecy, outbound-only rule
9. ThingsBoard	Cloud service	Device, rule engine, dashboards	Telemetry > history, widgets, alarms	Store, visualise, alarm	Service/internet outage, wrong rule	Role-based users, token rotation,
10. People / process	Maintenance team	CMMS or log	Alarm > action	Inspect, diagnose, record	Alarm ignored or not acknowledged	Access roles

Prototype vs Actual Industrial Implementation

Aspect	Current Prototype	Actual Industrial Implementation
Source	Prosys Simulation Server	Real sensors / existing plant signals
PLC	Not used as physical source	Siemens PLC
Input	Simulated OPC UA values	Real sensor signals
Wiring	Not physically tested	Instrument cables / JBs / terminals
OPC UA	Prosys server	Siemens PLC OPC UA Server
Edge	Node-RED on development system	Industrial Edge Gateway
MQTT	Used	Used with industrial security
Cloud	ThingsBoard	ThingsBoard / approved cloud
Dashboard	Prototype	Production dashboard
Alarm	Demonstration logic	Commissioned plant alarm logic
Cybersecurity	Development setup	Industrial OT/IT security
Validation	Functional proof	FAT / SAT / commissioning

Future Scope: Toward Predictive Maintenance

What we have built is condition-based monitoring, not predictive maintenance. The next stage adds a machine-learning layer on top of the same data pipeline.



Planned ML applications, once historical plant data exists

1. Remaining Useful Life (RUL) estimation — forecast when a trend will cross a critical threshold, not just flag that it already has
2. Multi-parameter anomaly detection — catch abnormal combinations before any single threshold is breached.
3. Failure-type classification — once enough labeled historical failure events are recorded.

- Requires months of real plant operating and failure history — not buildable from the current prototype data

Final Conclusion

The project creates a bridge between physical agitator condition, the existing PLC/automation infrastructure and modern IIoT/cloud monitoring. A working prototype has proved the software data pipeline with simulated data. Real sensors, PLC integration, cable work, security approval and commissioning are the remaining engineering steps.

